Saif Eddine Nouma, PhD

Contact information

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Research Interests

My research interests include **applied cryptography, network security, IoT security, and machine learning**, with a focus on advancing the security and efficiency of emerging computing platforms. I am also interested in the security and efficiency of digital twins by using post-quantum cryptography, edge computing, and distributed designs.

Teaching Interests

Applied Cryptography, Information/Network Security, Malware Analysis, Machine Learning

Education

Doctor of Philosophy in **Computer Science** — Aug 2021-May 2026
University of South Florida, Tampa, FL, USA
Thesis: Lightweight and Resilient Cryptographic Protocols for Internet of Things (IoT)
Advisor: Attila Altay Yavuz

Bachelor of Science in **Computer Science** — Aug 2017-Dec 2020
University of Carthage, Tunis, Tunisia
Thesis: Applications of Machine Learning in Networking and IoT
Advisor: Khalil Drira

Associate of Science in **Mathematics** — Aug 2015-Jun 2017
University of Monastir, Monastir, Tunisia

Employment History

Graduate Research Assistant, USF, Tampa, FL — Aug 2021-present

Graduate Teaching Assistant, USF, Tampa, FL — Aug 2021-present

System Administrator, USF, Tampa, FL — Aug 2023-Jun 2024

Software Engineer, Kopileft Services Inc., Tunis, Tunisia — Jul 2020-Jul 2021

ML Engineer Intern, LAAS-CNRS, Toulouse, France — Jan 2020-Jul 2020

ML Engineer Intern, Wevioo, Tunis, Tunisia — Jun 2019-Aug 2019

Patents

2. Attila A. Yavuz, **Saif E. Nouma**. System and Method for Cryptographic Forensic Audits on Lightweight IoT and Digital Archives. **US Patent** US20240007300A1, 2024.

1. Attila A. Yavuz, **Saif E. Nouma**. Hardware Supported Authentication and Signatures for Wireless, Distributed and Blockchain Systems. **US Patent** US20230308289A1, 2023.

Journal Publications

6. **Saif E. Nouma**, Attila A. Yavuz. Lightweight and High-Throughput Secure Logging for Internet of Things and Cold Cloud Continuum. *ACM Transactions on Internet of Things* ***(ACM TIoT)***, 2026. *(IF: 3.5)*

5. **Saif E. Nouma**, Attila A. Yavuz. Lightweight and Resilient Signatures for Cloud-Assisted Embedded IoT Systems. ***Security and Privacy***, Vol. 9, 2026. *(IF: 2.1)*

4. Attila A. Yavuz, Saleh Darzi, **Saif E. Nouma**. LiteQSign: Lightweight and Quantum-Safe Signatures for Heterogeneous IoT Applications. ***IEEE Access***, Vol. 13, pp. 171442-171456, 2025. (*IF: 3.6*)

3. Aaron Pendino, Nghia Nguyen, **Saif E. Nouma**, Jing Wang, Attila A. Yavuz, Yasin Yilmaz, Gokhan Mumcu. Additively Manufactured RF Electronics with Structurally Integrated Physically Unclonable Functions for Wireless System Security. ***IEEE Access***, Vol. 13, pp. 145042-145059, 2025. (*IF: 3.6*)

2. Kiarash Sedghighadikolaei, Attila A. Yavuz, **Saif E. Nouma**. Signer-Optimal Multiple-Time Post-Quantum Hash-Based Signature for Heterogeneous IoT Systems. ***Internet of Things***, Vol. 33, 101694, 2025. (*IF: 7.6*)

1. **Saif E. Nouma**, Attila A. Yavuz. Trustworthy and efficient digital twins in post-quantum era with hybrid hardware-assisted signatures. *ACM Transactions on Multimedia Computing, Communications, and Applications* ***(ACM TOMM)***, 2024. (*IF: 6.0*)

Conference Publications

7. **Saif E. Nouma**, Attila A. Yavuz. Lightweight and Breach-Resilient Authenticated Encryption Framework for Internet of Things. In *43rd IEEE Military Communications Conference* ***(IEEE MILCOM)***, October 2025, Los Angeles, CA.

6. **Saif E. Nouma**, Attila A. Yavuz. Lightweight Digital Signatures for Internet of Things: Current and Post-Quantum Trends and Visions. In *6th IEEE Conference on Dependable and Secure Computing* ***(IEEE DSC)***, May 2023, Tampa, FL.

5. **Saif E. Nouma**, Attila A. Yavuz. Practical Cryptographic Forensic Tools for Light-weight Internet of Things and Cold Storage Systems. In *8th ACM/IEEE Conference on Internet of Things Design and Implementation* ***(ACM/IEEE IoTDI)***, May 2023, San Antonio, TX.

4. **Saif E. Nouma**, Attila A. Yavuz. Post-Quantum Forward-Secure Signatures with Hardware-Support for Internet of Things. In *58th IEEE International Conference on Communications* ***(IEEE ICC)***, May 2023, Rome, Italy.

3. Attila A. Yavuz, Kiarash Sedghighadikolaei, Saleh Darzi, **Saif E. Nouma**. Beyond Basic Trust: Envisioning the Future of NextGen Networked Systems and Digital Signatures. In *5th IEEE Conference on Trust, Privacy, and Security in Intelligent Systems and Applications* ***(IEEE TPS-ISA)***, May 2023, Atlanta, GA.

2. Attila A. Yavuz, **Saif E. Nouma**, Thang Hoang, Duncan Earl, Scott Packard. Distributed Cyber infrastructures and Artificial Intelligence in Hybrid Post-Quantum Era. In *4th IEEE Conference on Trust, Privacy, and Security in Intelligent Systems and Applications* ***(IEEE TPS-ISA)***, December 2022, Virtual.

1. Attila A. Yavuz, Duncan Earl, Scott Packard, **Saif E. Nouma**. Hybrid Low-Cost Quantum-Safe Key Distribution. In ***Quantum 2.0*** – *Optica*, May 2022, Everett, MA.

E-prints

3. **Saif E. Nouma**, Gokhan Mumcu, Attila A. Yavuz. Diamond: Design and Implementation of Breach-Resilient Authenticated Encryption Framework for Internet of Things, arXiv preprint arXiv:2601.00353, 2026. Under review at *ACM Transactions on Internet of Things* ***(ACM TIoT)***. *(IF: 3.5)*

2. Saleh Darzi, **Saif E. Nouma**, Kiarash Sedghi, Attila A. Yavuz. QPADL: Post-Quantum Private Spectrum Access with Verified Location and DoS Resilience. *arXiv preprint arXiv:2510.03631*, 2025. Under review at *IEEE Transactions on Information Forensics and Security* ***(IEEE TIFS)***. *(IF: 8.0)*

1. **Saif E. Nouma**. Applications of Machine Learning (ML) in Networking and IoT. preprint hal-02932494, 2020.

Teaching Experience

Graduate Teaching Assistant

CIS 4212/6214: Privacy-Preserving and Trustworthy Cyber-Infrastructures	Spring 2026
COP 4538 IT Data Structures	Fall 2021

Guest Lecturer
CIS 4930/6930: Cryptography: Theory and Practice — Fall 2025
CIS 4212/6214: Privacy-Preserving and Trustworthy Cyber-Infrastructures — Spring 2024

Research Experience

Graduate Research Assistant — Aug 2021-present
Design and implement efficient authenticated encryption primitives, including:
- Graphene: Designed efficient and compromise-resilient authenticated encryption constructions based on the AES-GCM and Chacha20-Poly1305 standards, with full-fledged implementation on 32-bit ARM Cortex-M4, achieving orders-of-magnitude performance improvements and low-latency compared to the state-of-the-art counterparts, presented at IEEE MILCOM 25. My implementation is open-source at `https://github.com/saifnouma/graphene`
- Diamond: Improved the Graphene scheme by incorporating an efficient key update and more algorithmic formality with provable security. Extensive performance evaluation on embedded devices, 64-bit ARM Cortex-A72 and 8-bit AVR ATmega2560 confirms the efficiency of Diamond over Graphene and NIST lightweight cryptographic standard Ascon. My implementation is open-source at `https://github.com/saifnouma/diamond`

Design and implement efficient digital signature primitives for constrained devices, including:
- LRSHA: Designed and implemented efficient and resilient digital signatures by leveraging distributed cloud assistance and secure hardware Intel SGX. Extensive performance evaluation on 8-bit AVR ATmega2560 and commodity hardware confirms the efficiency of LRSHA. My implementation is open-source at `https://github.com/saifnouma/lrsha`
- POSLO: Designed an efficient secure logging protocol, achieving extended battery longevity on constrained devices compared to BLS. Developed a novel GPU acceleration for batch verification of large log entries on the cloud side, resulting in significant speedup compared to baseline CPU execution. Our work resulted in filing one U.S. patent and publications at ACM/IEEE IoTDI 23 and ACM TIoT 25. My implementation is open-source at `https://github.com/saifnouma/poslo`
- HYHASES: Designed efficient, forward-secure, and post-quantum digital signatures with TEE hardware support, implemented on 8-bit AVR ATmega2560 and Intel SGX, achieving 34× better efficiency than the NIST ML-DSA standard, presented at IEEE ICC 23. Later, I extended this work to a hybrid signature, leveraging a conventional aggregate signature to significantly reduce bandwidth overhead and energy usage at low-end devices. The extended work is published in ACM TOMM 2024. My implementation is open-source at `https://github.com/saifnouma/hyhases`

Professional Experience

System Administrator, USF, Tampa, FL — Aug 2023-Jun 2024
- Developed, configured, and maintained real-time dashboards and alert rules using Prometheus and Grafana stacks, tracking I/O network throughput and node/container health across an HPC cluster and the machines of the college infrastructure
- Developed Ansible roles and playbooks to support continuous integration and task automation

Software Engineer, Kopileft Services Inc. — Jul 2020-Jul 2021
- Improved e-commerce backends by migrating Java-based SOAP web services to Kotlin, developed services on Apache Tomcat, and implemented Exposed-based data-access layers with PostgreSQL
- Implemented the BI reporting and dashboard infrastructure by optimizing ≈ 50% of SQL queries and stored procedures written in PostgreSQL, improving data freshness and reliability

ML Engineer Intern, LAAS-CNRS, Toulouse, France — Jan 2020-Jul 2020
- Implemented and benchmarked LSTM/GRU models using TensorFlow and scikit-learn on a real-world network traffic (1.6 TB capture), achieving 3% MAPE and halving error compared to SVR. My implementation is open-source at https://github.com/SaifNOUMA/Network-Traffic-Prediction
- Implemented a distributed early-exit CNN model for edge IoT networks using TensorFlow, containerized with Docker, and performed communication using ZeroMQ, enabling faster inference. My implementation is open-source at https://github.com/SaifNOUMA/Edge-Computing/tree/master/dnn_partitioning
- Developed and implemented a DQN algorithm for dynamic task offloading and resource allocation in multi-user edge computing, using ns3-gym, achieving improved latency-energy tradeoffs. My implementation is open-source at https://github.com/SaifNOUMA/Edge-Computing/tree/master/task_offloading_optimization

ML Engineer Intern, Wevioo, Tunis, Tunisia Jun 2019-Aug 2019

- Developed, trained, and fine-tuned a Siamese CNN (S-CNN) with contrastive loss on a collected real-world dataset of genuine/forged handwritten signature pairs for fake signature detection
- Collaborated with computer-vision team on signature extraction from handwritten bank checks and deployed the prototype to local banks, which served as a major step towards mitigating fraud

SKILLS

Programming Languages: C/C++, Assembly, Python, Java, Kotlin, MATLAB, CUDA

Security & Cryptography: OpenSSL, WolfSSL, Mbed-TLS, MITRE ATT&CK, SGX

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib

DevOps: Git, SVN, Docker, Ansible, Prometheus, Grafana, Nagios, Slurm

Database: PostgreSQL, MySQL

Embedded Hardware: ARM Cortex-M4, ARM Cortex-A72, 8-bit AVR ATMega series

HONORS & AWARDS

NSF/USF International Travel Grant for IEEE ICC 2023 ($3,000) 2023

Top 0.5% (Rank 4/800) at Tunisian National Engineering Entrance Exam 2017

PUBLIC TALKS

Research paper presentation at IEEE MILCOM, Los Angeles, CA, USA 2025

Ph.D. major area Presentation at Tampa, FL, USA 2024

Research paper presentation at IEEE DSC, Tampa, FL, USA *2023*

Research paper presentation at ACM/IEEE IoTDI, San Antonio, TX, USA 2023

SERVICES

Reviewer

IEEE Transactions on Dependable and Secure Computing (IEEE TDSC) 2026

Blockchain: Research and Applications (Elsevier), Computer Networks (Elsevier), Reliability Engineering & System Safety (Elsevier), Security and Privacy (Wiley) 2025

IEEE Transactions on Information Forensics and Security (IEEE TIFS) 2024

REFERENCES

Attila Altay Yavuz, Ph.D.
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